

Sarathi — Product Requirements Document

Version: 2.0 | **Date:** April 2026 | **Status:** Draft — Google Cloud Edition

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1. Product Overview

1.1 Vision

Sarathi is a personal lifecycle AI assistant that guides you from the moment your alarm rings in the morning to the moment you fall asleep at night. Named after the charioteer — as Krishna was Arjuna's Sarathi in the Mahabharata — the product acts as an intelligent, proactive guide that knows your goals, understands your patterns, tracks your progress, and holds you accountable.

Sarathi is not a chatbot. It is not a reminder app. It is an always-on, context-aware companion that learns who you are over time and actively steers your life toward the goals you set for yourself.

The entire product runs on **Google Cloud**. One GCP project, one billing account, all credits used from a single place.

1.2 Name & Meaning

Sarathi (■■■■■) means charioteer in Sanskrit. As Krishna guided Arjuna through the chaos of battle in the Mahabharata, Sarathi guides the user through the chaos of daily life — keeping them focused, accountable, and moving toward their goals. Sarathi knows your strengths and covers your weaknesses. If you forget, it remembers. If you get distracted, it pulls you back. If you achieve something, it acknowledges it honestly.

1.3 The Core Problem

People — especially students and young professionals — have the talent and the intention to succeed but struggle with discipline, memory, and consistency. They forget tasks, get distracted, lose track of time, and have no system connecting daily actions to long-term goals. Existing tools solve individual fragments (reminders, notes, habit trackers) but none act as a unified, intelligent, personalized life guide.

2. Problem Statement

The target user has talent but lacks discipline. Specifically:

- Forgets things he needs to do
- Gets distracted easily and loses track of work sessions
- Has no feedback loop connecting daily actions to long-term goals
- Lacks a system that proactively nudges and guides rather than waiting to be opened
- Needs something that learns his patterns and improves guidance over time

No existing app solves all of this in one place. Sarathi does.

3. Target User

Phase 1 — Single User (MVP)

The initial user is the product creator himself: a college student with ambitious goals, irregular routines, and a tendency to forget or get distracted. Building for one person first ensures the product actually solves a real problem before scaling.

Phase 2 — Students

After validating the product personally, Sarathi opens to students across Indian colleges facing the same discipline and focus challenges.

Phase 3 — Professionals

Once the student use case is proven, Sarathi expands to working professionals who need the same lifecycle accountability but in a corporate context.

4. Core Design Principles

- **Proactive over Reactive** — Sarathi reaches out to you; you do not always have to open the app
 - **Personalized over Generic** — Every interaction is shaped by your history, goals, and learned patterns
 - **Accountability over Comfort** — Sarathi tells you what you need to hear, not what you want to hear. No pampering. No hollow motivation.
 - **Privacy First** — Raw screen data and voice audio never leave the device. Only summarized habit text goes to cloud.
 - **Lifecycle Aware** — Sarathi understands your full day as a connected arc, not isolated events
 - **Google Cloud Native** — All cloud services run on GCP. One platform, one billing account, credits cover everything.
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5. Complete User Lifecycle

5.1 Morning — Wake Up & Briefing

When the user's alarm time arrives, Sarathi monitors whether the alarm has been dismissed. If the user misses or snoozes it, Sarathi proactively opens on the phone screen with a motivational wake-up prompt generated by **Gemini 2.5 Pro on Vertex AI** — tied directly to the user's actual goals, not generic quotes. The tone is firm and direct. No emotional coddling.

After the user confirms they are awake, Sarathi delivers a personalized morning briefing:

- Top goals for the day
- Workout plan if it is a gym day
- Breakfast suggestion based on dietary goals
- Time-to-leave countdown powered by **Google Maps Routes API** using home location and live traffic
- Real-time alerts so the user is never late

5.2 During the Day — Quick Capture

While at college or elsewhere, the user can quickly capture tasks, thoughts, or information in three modes:

- **Voice** — Transcribed by **Cloud Speech-to-Text (Chirp)**, accurate in Hindi and English
- **Text** — Typed directly into the app
- **Image** — Photographed via CameraX, uploaded to **Google Cloud Storage (GCS)**

Each capture is auto-categorized by **Gemini 2.5 Flash-Lite** and embedded into **Vertex AI Matching Engine** for retrieval later that evening.

5.3 Evening — Homecoming & Debrief

Google Maps Geofencing API detects when the user arrives home. **Firebase Cloud Messaging** sends a push notification. The user taps it to open the evening debrief conversation.

Sarathi initiates a natural language conversation powered by **Gemini 2.5 Pro with RAG context** from Vertex AI Matching Engine:

- Surfaces specific captures from the day
- Reviews morning goals by name — what was achieved, what was not
- Summarizes the day
- Plans the evening work session with specific tasks and time estimates

5.4 Evening — Laptop Focus Session

When the user switches to their laptop, the Sarathi web app activates screen monitoring:

- Screen captured every 30 seconds via browser Screen Capture API
- In MVP: frame classified by **Gemini 2.5 Flash-Lite** as PRODUCTIVE or DISTRACTION
- In Phase 2: frame classified locally by **Gemma 7B via Ollama** — never sent to cloud
- If DISTRACTION detected for more than 5 consecutive minutes: firm nudge overlay appears referencing the user's actual current goal
- Focus time and distraction events logged to **Cloud SQL** as summarized habit text
- Raw frames discarded immediately — never stored

5.5 Night — Debrief & Alarm Setup

When Sarathi detects the user is heading to sleep (time pattern + schedule), it sends an FCM notification. The night debrief covers:

- What was achieved today
- What went wrong — specific and honest
- One key improvement area identified by Sarathi from the full day's data
- Sarathi asks for tomorrow's wake time, suggests optimal time based on sleep goals
- Alarm set in Android system via **AlarmManager**
- Full day summary embedded into Vertex AI Matching Engine for long-term memory

6. Google Cloud Technology Stack

6.1 Why Google Cloud

All Sarathi services run on Google Cloud Platform (GCP), billed to existing Google Cloud credits. No separate subscriptions. No multiple billing accounts. One platform covers the LLM brain, database, cache, vector search, file storage, push notifications, auth, geofencing, CI/CD, and monitoring.

6.2 Complete GCP Services Map

Sarathi Need	Google Cloud Service	Notes
Cloud LLM — main conversations	Vertex AI Gemini 2.5 Pro	Morning, evening, night debrief
Cloud LLM — quick tasks	Vertex AI Gemini 2.5 Flash-Lite	Capture categorization, screen classification (MVP)
Real-time voice AI	Gemini Live API (Phase 2+)	Low-latency bidirectional voice
Text embeddings for RAG	Vertex AI text-embedding-004	768-dim vectors
Vector similarity search	Vertex AI Matching Engine	Replaces Pinecone entirely
Voice transcription	Cloud Speech-to-Text (Chirp)	125 languages, Hindi + English
Text-to-speech (optional)	Cloud Text-to-Speech	Sarathi speaking back
Backend hosting	Cloud Run (FastAPI)	Serverless, pay per request
Primary database	Cloud SQL (PostgreSQL 15)	Managed, no server ops
Session cache	Memorystore (Redis 7)	Conversation context, rate limiting
File storage	Cloud Storage (GCS)	Images, model files, training data
Push notifications	Firebase Cloud Messaging	Free, deeply Android-integrated
User authentication	Firebase Authentication	Free, handles JWT automatically
Crash reporting	Firebase Crashlytics	Free
Geofencing + routing	Google Maps Platform	Geofencing API + Routes API + Places API
Secrets & credentials	Secret Manager	All API keys, DB passwords
Encryption keys	Cloud KMS	Per-user AES-256 key management
CI/CD pipeline	Cloud Build + Artifact Registry	Trigger on GitHub push to main
Monitoring & logs	Cloud Logging + Cloud Monitoring	Dashboards, alerts, error tracking
Scheduled jobs	Cloud Scheduler + Cloud Run Jobs	Fine-tuning trigger (Phase 3)
Fine-tuning training	Vertex AI Training + DGX A100	Phase 3 — personalized model updates
On-device Android	MediaPipe + TensorFlow Lite	Gemma inference on phone (Phase 2)
On-device desktop	Ollama	Gemma 7B local inference (Phase 2)

6.3 Three-Tier Architecture

- TIER 1 – LOCAL (Privacy Layer)
Android: MediaPipe + TFLite (Gemma 2B) [Phase 2]
Desktop: Ollama (Gemma 7B) [Phase 2]
MVP: local processing skipped, Gemini Flash-Lite used directly
- TIER 2 – GOOGLE CLOUD MEMORY
Cloud SQL (PostgreSQL 15) – structured data
Memorystore (Redis 7) – session cache
Vertex AI Matching Engine – vector memory (RAG)
Cloud Storage (GCS) – files and model checkpoints
- TIER 3 – GOOGLE CLOUD BRAIN
Vertex AI Gemini 2.5 Pro – main reasoning and conversation
Vertex AI Gemini Flash-Lite – fast, cheap micro-tasks
Cloud Speech-to-Text – voice transcription
Google Maps Platform – location and routing

6.4 Estimated Monthly Credit Usage (MVP Phase)

Service	Estimated Monthly Cost
Cloud Run (FastAPI backend)	~\$5–15
Cloud SQL (PostgreSQL, db-f1-micro)	~\$10
Memorystore (Redis 7, 1GB Basic)	~\$25
Vertex AI Gemini 2.5 Pro (conversations)	~\$20–50
Vertex AI Gemini Flash-Lite (micro-tasks)	~\$3–8
Vertex AI Matching Engine	~\$10
Cloud Storage	~\$1–3
Cloud Speech-to-Text (Chirp)	~\$2–5
Firebase Auth + FCM	Free
Google Maps Platform	Free tier (\$200/month credit)
Cloud Build	Free tier (120 min/day)
Secret Manager	~\$1
Total MVP estimate	~\$77–127/month

Google Cloud credits cover this comfortably during all of development and testing.

7. Feature List

F01 — User Onboarding & Profile Setup

First-time setup where the user registers via Firebase Authentication, sets goals, daily schedule, and home location. All data seeded into Cloud SQL and Vertex AI Matching Engine.

Key requirements:

- Firebase Authentication (email + password)
 - Up to 10 goals with title, description, and priority
 - Daily schedule: wake time, gym time, college leave time, sleep target
 - Home location via Google Places API autocomplete
 - All data embedded via text-embedding-004 into Matching Engine on save
 - Completable in under 5 minutes
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F02 — Morning Wake-Up & Alarm Monitor

Sarathi monitors alarm time. If missed or snoozed, the app opens fullscreen with a Gemini-generated motivational message tied to the user's actual goals. Firm. No hollow quotes.

Key requirements:

- Alarm time set inside app — stored in Cloud SQL + native AlarmManager
 - WorkManager background job checks alarm dismissal
 - FCM wake-up trigger if alarm missed
 - Gemini 2.5 Pro generates motivational message using Matching Engine context (goals)
 - User taps "I'm awake" to proceed — cannot skip
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F03 — Morning Briefing

After wake-up, Sarathi delivers a personalized briefing using Gemini 2.5 Pro with top-8 RAG results from Vertex AI Matching Engine.

Key requirements:

- Top 3 goals for today
 - Workout plan if gym day
 - Breakfast suggestion
 - Time-to-leave via Google Maps Routes API with live traffic
 - Live countdown timer on screen
 - Conversational follow-up questions supported (Gemini streaming)
 - Loads within 5 seconds
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F04 — Quick Capture

Three input modes for capturing tasks, thoughts, and observations during the day.

Key requirements:

- Voice: Android SpeechRecognizer (short) + Cloud Speech-to-Text Chirp (long)
 - Text: direct input field
 - Image: CameraX photo → uploaded to GCS bucket → signed URL stored in Cloud SQL
 - Auto-categorized by Gemini 2.5 Flash-Lite: task / reminder / observation / question
 - Each capture embedded into Vertex AI Matching Engine
 - Capture under 10 seconds from open to saved
 - Persistent floating action button in app
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F05 — Homecoming Detection & Trigger

Google Maps Geofencing API detects the user arriving home. Firebase Cloud Messaging sends a push notification opening the evening debrief.

Key requirements:

- Home location set via Google Places API autocomplete
 - 100m radius geofence via Google Maps Geofencing API
 - Geofence ENTER transition triggers FCM push + POST /lifecycle/homecoming
 - Fires only when lifecycle phase in Memorystore is day
 - Manual fallback trigger in Settings
 - Geofence resets each morning
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F06 — Evening Debrief Conversation

Natural language debrief powered by Gemini 2.5 Pro with full RAG context from Matching Engine.

Key requirements:

- Context-aware opener referencing something specific from the day
 - Surfaces today's captures from Cloud SQL during conversation
 - Reviews morning goals by name — checks progress on each
 - Streamed via WebSocket — no waiting for full response
 - Voice or text input supported
 - Evening work plan generated after debrief
 - Conversation persisted in Memorystore (active) + Cloud SQL (long-term)
 - Turns embedded into Vertex AI Matching Engine
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F07 — Laptop Screen Monitoring

Desktop web app monitors screen activity. Distractions flagged with goal-tied nudge. Focus session data synced to Cloud SQL as habit summaries.

Key requirements (MVP):

- Browser Screen Capture API captures frame every 30 seconds
- Frame classified by Gemini 2.5 Flash-Lite as PRODUCTIVE or DISTRACTION
- 5-minute distraction threshold before nudge appears
- Nudge message references user's actual current goal by name
- Session summary: total focus time, distraction count, longest streak
- Habit summary synced to Cloud SQL + embedded into Matching Engine
- Raw frames never stored or sent to any database

Phase 2 upgrade: Frame classified locally by Gemma 7B via Ollama — frames never leave the device.

F08 — Night Debrief & Alarm Setup

Before sleep, Sarathi conducts a full day review using Gemini 2.5 Pro with complete RAG context. Sets next morning's alarm through conversation.

Key requirements:

- Sleep time detected via schedule config + Redis time pattern
 - FCM push: "Ready to wrap up the day?"
 - Night debrief covers: achievements, what went wrong, one specific improvement area
 - Sarathi feedback references actual events from the day — not generic
 - Alarm time extracted from conversation via Gemini structured output
 - Alarm set in Android system via AlarmManager + stored in Cloud SQL
 - Full day summary embedded into Vertex AI Matching Engine (long-term memory)
 - Lifecycle phase reset to morning in Memorystore after alarm set
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F09 — General Conversational Chat

At any time, the user can open Sarathi for a natural conversation. Gemini 2.5 Pro uses full RAG context. User can update goals or add captures by talking naturally.

Key requirements:

- Available from main navigation at all times
 - Full Matching Engine RAG context in every response
 - Natural language goal and capture updates: extracted by Gemini, saved to Cloud SQL
 - Conversation history persists across app restarts (Cloud SQL)
 - Streaming response — first tokens within 2 seconds
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F10 — Settings & Profile Management

User can view and update all profile data. All changes immediately re-embedded into Vertex AI Matching Engine.

Key requirements:

- Edit goals, schedule, home location (Google Places autocomplete)
- Toggle screen monitoring on/off (desktop)
- View today's captures with resolved/unresolved status
- Clear conversation history (Cloud SQL + Matching Engine + Memorystore)
- All updates immediately reflected in RAG memory

8. Phased Build Plan

Pre-Sprint 0 — GCP Setup (2–3 days)

Before writing a single line of application code, provision all Google Cloud services:

- Create GCP project `sarathi-prod` and link Google Cloud credits
- Enable APIs: Vertex AI, Cloud Run, Cloud SQL, Memorystore, Cloud Storage, Speech-to-Text, Maps Platform, Secret Manager, Cloud Build, Artifact Registry
- Create Firebase project, link to GCP project, enable Auth and FCM
- Create Cloud SQL instance (PostgreSQL 15, db-f1-micro, us-central1)
- Create Memorystore Redis (1GB, Basic tier, us-central1)
- Create GCS bucket `sarathi-app`
- Create Vertex AI Matching Engine index (768-dim, cosine similarity)
- Add all secrets to Secret Manager (DB password, Maps key, etc.)
- Create Cloud Run service account with correct IAM roles
- Connect GitHub repo to Cloud Build trigger

Phase 1 — MVP (8 Sprints, ~8 Weeks)

Goal: Full lifecycle working for one user. All cloud services on GCP. Validate that Sarathi actually solves the discipline problem.

Sprint	Focus	Key Deliverable
Sprint 0	GCP Setup	All services provisioned, Firebase configured
Sprint 1	Backend + Auth + RAG	Cloud Run API live, Cloud SQL schema, Matching Engine working
Sprint 2	Gemini Chat	Full conversation with RAG context streaming on web UI

Sprint	Focus	Key Deliverable
Sprint 3	Morning Flow (Android)	Alarm + wake-up + briefing + Maps Routes time-to-leave
Sprint 4	Quick Capture (Android)	Voice (Chirp) + text + image (GCS) all syncing
Sprint 5	Homecoming + Debrief	Geofence trigger → FCM → evening debrief conversation
Sprint 6	Screen Monitoring (Web)	Gemini-powered distraction detection + nudge
Sprint 7	Night Debrief + Alarm	Honest debrief + AlarmManager set through conversation
Sprint 8	Polish + E2E Testing	Full day works without bugs, Cloud Monitoring live

Phase 2 — Local Model Integration

Goal: Replace Gemini API calls for real-time tasks with local Gemma models. Improve privacy.

- Integrate Ollama on desktop — route screen monitoring to local Gemma 7B
- Integrate MediaPipe TFLite Gemma 2B on Android
- Build privacy gateway: local model decides what to send to Vertex AI
- Model files stored in GCS, pulled by device automatically

Phase 3 — Self-Learning & Fine-Tuning

Goal: Sarathi becomes genuinely personalized through periodic fine-tuning.

- Cloud Scheduler triggers weekly training job
- Data pipeline: export anonymized habit summaries + conversations from Cloud SQL as JSONL
- Upload to GCS training data bucket
- Fine-tuning via LoRA on Gemma 2B/7B using Vertex AI Training or DGX A100 cluster
- Output: GGUF (Ollama desktop) + TFLite (Android MediaPipe)
- Push updated model checkpoints to GCS
- Device pulls silently on Wi-Fi + charging

Phase 4 — Scale & Product

Goal: Open Sarathi to other students, then professionals.

- Multi-user backend hardening, rate limiting, billing
 - Onboarding flow for new users
 - Gemini Live API for real-time voice conversations
 - Analytics dashboard for users to review their own patterns
 - Expand fine-tuning pipeline for multiple concurrent users
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9. Data & Privacy Architecture

9.1 What Stays Local (Never Sent to Cloud)

- Raw screen frames — processed locally (Phase 2) or discarded immediately after Gemini classification (MVP)
- Voice audio — transcribed by Chirp, raw audio discarded immediately
- Full conversation audio — text only stored, audio discarded

9.2 What Goes to Google Cloud

- Summarized habit patterns (text only, e.g. "User distracted by YouTube 3x during session")
- Task captures (text + GCS image URLs)
- Goals and schedule data
- Conversation text (after local model approves — Phase 2; directly in MVP)
- Anonymized behavioral embeddings for Matching Engine

9.3 Google Cloud Security Services

Need	GCP Service
Credential storage	Secret Manager
Per-user encryption keys	Cloud KMS
User authentication	Firebase Authentication
Backend token verification	Firebase Admin SDK
Data in transit	TLS 1.3 (automatic on Cloud Run + all GCP APIs)
Data at rest	AES-256 (default on Cloud SQL, GCS, Memorystore)
Access control	Cloud IAM (least privilege per service account)

9.4 Data Classification Table

Data Type	Stored Where	Encrypted	Sent to Cloud?
Raw screen frames	Nowhere (discarded after classification)	N/A	Never
Voice audio	Nowhere (transcribed, discarded)	N/A	Never
Habit summaries (text)	Cloud SQL	AES-256	Yes
Task captures	Cloud SQL + Matching Engine	AES-256	Yes
Goals	Cloud SQL	AES-256	Yes
Conversation text	Memorystore + Cloud SQL	AES-256	Yes
Image captures	Google Cloud Storage	GCS server-side	URL only in DB
Model checkpoints	Google Cloud Storage	GCS server-side	Download only

10. Key Challenges & Mitigations

Challenge	Risk	Mitigation
Vertex AI Gemini quota limits during testing	Medium / High	Request quota increase early; use Flash-Lite for non-critical tasks
Matching Engine cold start latency on first query	High / Medium	Warm up index on app start; cache frequent queries in Memorystore
Android Geofence unreliable indoors	High / Medium	Manual fallback trigger in Settings; test on target device
FCM delivery delay on Indian OEMs (Xiaomi, Realme)	High / Medium	Use high-priority FCM messages; test on physical device
Screen Capture API blocked by browser	Low / High	Test on Chrome first; Electron wrapper documented as Phase 2 fallback
Gemini Flash-Lite screen classification latency	Medium / Medium	Compress frames to quality 0.4; Flash-Lite is fast enough at 30s intervals
GCP credits running out during heavy development	Low / High	Weekly billing dashboard check; set budget alert at 50% of credits
Cloud SQL connection limits on Cloud Run	Medium / Medium	Use Cloud SQL connector with SQLAlchemy connection pooling

Challenge	Risk	Mitigation
Scope creep delaying MVP	High / High	Hard feature freeze after Sprint 0; no new features until MVP ships
Local fine-tuning data privacy	Medium / High	All training data anonymized and encrypted in GCS before processing

11. Success Metrics

Phase 1 MVP — Personal Prototype

Metric	Target
Daily usage	Creator uses Sarathi every day for 2 consecutive weeks
Morning briefing	Feels personal and motivating, references actual goals
Capture surfacing	80%+ of day captures appear in evening debrief
Distraction detection	Flags distractions within 30 seconds of threshold
Night alarm	Set through Sarathi at least 10 out of 14 nights
Self-reported improvement	Creator reports measurable improvement in discipline
Infrastructure cost	All costs covered by Google Cloud credits

Phase 4 — Product at Scale

Metric	Target
Day 30 retention	Above 60% for active users
Personalization feel	Users report Sarathi "knows them" within 2 weeks
Daily interactions	Average 5+ meaningful interactions per day
Net Promoter Score	Above 50

12. Out of Scope

Feature	Phase
Local Gemma models (MediaPipe / Ollama)	Phase 2

Feature	Phase
Gemini Live API voice sessions	Phase 2+
Fine-tuning pipeline (Vertex AI Training / DGX A100)	Phase 3
Multi-user backend	Phase 4
Native desktop Electron app	Phase 4
College system integrations (timetable, attendance)	Removed from scope
Cloud Text-to-Speech (Sarathi voice)	Phase 2+
Analytics dashboard for users	Phase 4
Multi-language UI	Phase 4
Social or sharing features	Not planned

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